

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of M. Pharm Pharmaceutics (Drug Regulatory Affairs)

University Theory Examination April/May 2015

PH820.06 Drug Regulatory Affairs III

Date: 13.05.2015 ,Wednesday Time:10.00 a.m to 1.00 p.m Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

Q.1 A) What are Biologics? Explain approval process of Biologics in brief as per US FDA. **10**

Q.2 A) Discuss in detail SUPAC guidelines for immediate release solid oral dosage form. **10**

- I. Components and composition
- II. Site changes
- III. Changes in batch size (scale-up/scale-down)
- IV. Manufacturing

OR

Q.2 B) Classify Medical Devices based on Risk and write down in brief about its approval process as per US-FDA **10**

Q.3 Answer the following (Any three) **15**

- (A) Paragraph Certification of ANDA (Para I- ParaIV)
- (B) Short Note:INDA
- (C) Orange Book
- (D) Regulation of OTC Product in USA.

SECTION-II

Q.4 A) What is NDA? How it is differ from ANDA? Explain different sections of NDA as per US FDA. **10**

Q.5 A) Regulatory considerations for manufacturing, packaging and labeling of pharmaceuticals in EU. **10**

OR

Q.5 B) Compare the Drug Master File Procedure as per USA and Europe in brief. **10**

Q.6 Answer the following (Any three) **15**

(A) Orphan Drug Regulation as per European Union.

(B) Regulatory pathway for drug approval in Europe in brief: Centralized and Decentralized concept.

(C) Regulation of Herbal formulation in Europe.

(D) Regulatory requirement for conducting clinical trials in Europe.